Archit Pasayat

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EDUCATION

Indian Institute of Technology, Bhilai — Durg, Chhattisgarh
Bachelor of Technology in Electrical Engineering — Jul 2026 – Present

National Institute of Technology, Durgapur — Paschim Bardhaman, West Bengal
Bachelor of Technology in Electrical Engineering, CGPA: 8.97 — Aug 2025 – May 2026
Relevant Coursework: Data Structures and Algorithms, Basic Electrical and Electronics Engineering

EXPERIENCE

OpenLake IIT Bhilai — Durg, Chhattisgarh
Design and PR Member — Aug 2026 – Present

- Serve on the core organizing committee for Major League Hacking (MLH) events at IIT Bhilai, coordinating logistics, track planning, and execution for 150+ hackathon participants.

HBCSE, Tata Institute of Fundamental Research (TIFR) — Mumbai, Maharashtra
National Initiative on Undergraduate Science (NIUS) - Physics Scholar — Jun 2026 – Jul 2026

- Selected nationally among top 70 undergraduate science and engineering students for a prestigious multi-year research initiative.
- Participated in intensive theoretical lectures and conducted hands-on experimental laboratory research mentored by senior scientists.
- Formulated rigorous experimental setups in advanced physics, analyzing and modeling empirical data to study complex physical phenomena.

IEEE Student Branch NIT Durgapur — Paschim Bardhaman, West Bengal
Junior Coordinator — Feb 2025 – Jul 2026

- Contributed to the planning and execution of technical workshops, guest lectures, and student competitions, successfully engaging over 200+ student attendees.

ACADEMIC PROJECTS

Cyclone Forecast Lab | *Python, Machine Learning, JavaScript, XGBoost, APIs* — 2026

- Engineered a global machine learning pipeline utilizing the NOAA IBTrACS dataset to predict tropical cyclone tracks and intensities across seven ocean basins for 6- to 72-hour lead times.
- Evaluated diverse ML architectures (tree ensembles, boosting, neural networks) on post-2016 data, achieving an 18% reduction in 48-hour track prediction error over a persistence baseline.
- Deployed a serverless web application featuring live forecasts, historical backtests, and interactive performance metrics, executing trained models directly in the browser via JavaScript.

Network Intrusion Detection System | *Python, Machine Learning, Scikit-Learn, Pandas* — 2025

- Engineered an end-to-end classification pipeline on the CIC-IoT dataset to identify malicious network intrusions versus benign traffic.
- Achieved 99.4% overall accuracy on the highly imbalanced dataset, utilizing Macro F1-scoring (0.72) to rigorously evaluate minority class detection limits.
- Secured 2nd Place (Runner-up) in Epoch, a machine learning hackathon hosted by the IEEE Student Branch, NIT Durgapur.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, LaTeX, SQL, JavaScript
Data Analysis & ML: NumPy, Pandas, Scikit-Learn, XGBoost, PyTorch, TensorFlow
Tools & Platforms: Git, Linux (Ubuntu), Docker, AWS, MATLAB

AWARDS & HONORS

- Secured Top 1% ranking in JEE Advanced 2026 (AIR $\sim$10,000) and 98.6 percentile in JEE Main 2025 among 1.4+ million candidates.
- **Smart India Hackathon (SIH) 2026:** Selected in the campus internal hackathon phase, securing institutional nomination to advance to the national round.